TEXORA · SESSIONAL NOTES · STUDY SHEET

Production of Different Types of Seams

Sewing and specifying the four working seam constructions — superimposed, lapped/felled, bound and flat — as mounted samples, each labelled with its ISO 4916 class, stitch type and garment application.

TE-3108 · APPAREL MANUFACTURING TECHNOLOGY-II SESSIONAL · STUDY 1 · LEVEL 3 TERM 1

1. Aim and place of the study

Aim: to produce, identify and specify the standard seam types of ISO 4916 as sewn samples: superimposed, lapped (felled), bound and flat seams, each with its correct stitch type, allowance and application on shirt and trouser work.

2. Materials, machine settings and specimens

Materials and machines: single-needle lockstitch machine (stitch 301) set at 4 stitches/cm, needle Nm 80 sharp, core-spun thread tex 24; overlock machine (504) for the flat-seam sample; calico/poplin pieces 20 × 10 cm (two per seam), binding strip 3 cm wide, ruler, seam gauge, scissors and the sample card for mounting. **Machine settings to record before sewing:** stitch density 4/cm, needle and bobbin tensions balanced on a test piece (knot buried mid-ply), presser pressure medium.

Trouser measurement chart used throughout TE-3108 (men's basic trouser, garment values in cm): waist S 77 · **M 82** · L 87 · XL 92 (grade 5); hip 97 · **102** · 107 · 112 (grade 5); outseam (incl. waistband) 102 · **104** · 106 · 107; inseam 76 · **78** · 80 · 81; thigh 60 · **62** · 64 · 66 (grade 2); knee 44 · **46** · 48 · 50 (grade 2); hem (bottom) width 38 · **40** · 42 · 44 (grade 2); front rise 25 · **26** · 27 · 28. Size M is the working size unless the study states otherwise. Seam allowances, as in TE-2207/TE-2208: **1 cm** general, **1.5 cm** at side seams, waistband and other adjustment seams, **2.5 cm** at hems. The shirt values of TE-2207/2208 (size M chest 107, length 73, shoulder 46, sleeve 61.5, collar 40; grades 5/2/2/1.5/2 cm) are used unchanged wherever a shirt is the specimen.

3. The seams to produce

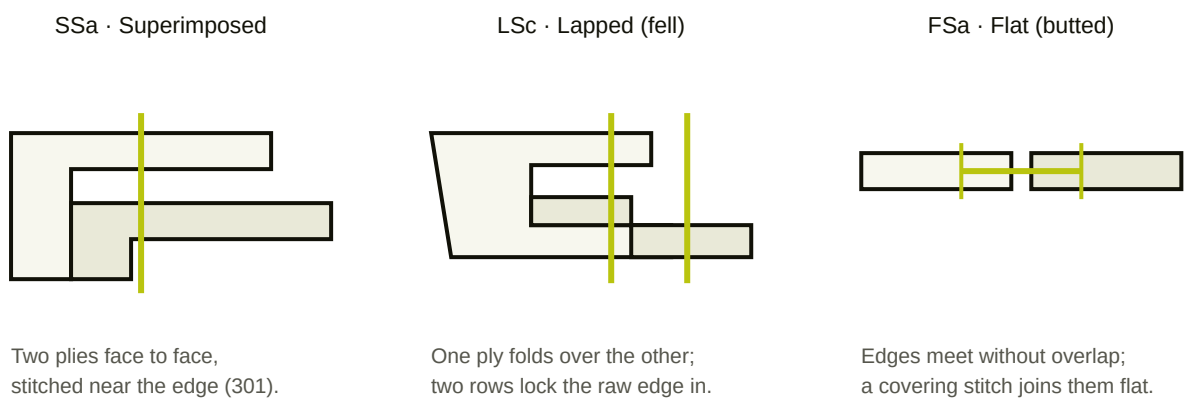


Figure 1 — Three seam constructions in cross-section (ISO 4916): superimposed (Class 1), lapped (Class 2) and flat (Class 4). Green lines are the stitching rows.

SAMPLE	ISO 4916 CLASS/TYPE	CONSTRUCTION (ALLOWANCE 1 CM UNLESS STATED)	GARMENT APPLICATION
A	Class 1 — SSa	Plies face to face, one row of 301 at the allowance line; press open, then a second specimen pressed to one side with edges overlocked (SSa + edge finish).	Shirt and trouser plain seams generally.
B	Class 2 — LSc (felled)	Plies wrong sides together, stitch; trim the lower allowance to 0.4 cm, fold the upper allowance over it, press and topstitch a second row of 301.	Shirt side seam, trouser inseam, jeans seams.
C	Class 3 — BSa (bound)	Binding strip folded over the raw edge of a single ply and stitched through all layers with 301.	Neckline and armhole binding, edge finishing.
D	Class 4 — FSa (flat)	Edges butted on a strip or overlapped flat and joined with 504/607 so no ridge stands on either face.	Knit underwear and sportswear seams.

4. Procedure

1. Sew a tension test piece; adjust until the 301 knot is invisible on both faces. Record density by counting stitches over 3 cm (must read 12).
2. Produce sample A twice (pressed open; pressed aside with overlocked edges). Measure the finished allowance from the needle line: 1.0 cm ± 0.1.
3. Produce sample B: check after trimming that the enclosed raw edge cannot escape the fold; the second row must catch the fold along its whole length — run a fingertip test and a pull test.

4. Produce sample C: the binding must show equal width on both faces (± 0.1 cm); corners are not required in this study.
5. Produce sample D on the overlock/flatlock: the join must lie flat with no ridge; stretch the sample by hand — it must extend without the threads cracking.
6. Mount each sample on the card; label with ISO class and type, stitch type, density, needle and thread, and one application.

Observation points — what to record

- Stitch balance of each sample (knot position) and density counted, not assumed.
- For the felled seam: whether the trimmed edge is fully enclosed along the whole length.
- Bound sample: binding width equality face-to-back, and stitch catching both folds.
- Flat sample: ridge height by fingertip, and extension before thread crack, estimated in percent.
- Any pucker, ply shift or skipped stitch, with the cause you assign (TE-3107 Topic 4 method).

Precautions

- Change the needle before the study; a session of samples sewn with a damaged needle teaches fault diagnosis, not seams.
- Sew test pieces of the actual sample fabric after every adjustment — settings belong to the fabric–machine pair.
- Keep fingers behind the needle line; samples are guided with the seam guide or a magnetic guide, never by fingertips at the needle.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Distinguish SSa and LSc by ply arrangement and give one garment position for each.
2. In sample B, why is the lower allowance trimmed to 0.4 cm before folding?
3. Your bound sample shows 0.9 cm binding on the face and 0.5 cm underneath. What went wrong in feeding?
4. Which sample must stretch without cracking, and which stitch types achieve that?

Answer key

1. SSa (Class 1) stacks the plies face to face and stitches one row near the edge — plain seams such as a trouser outseam. LSc (Class 2) laps the plies and folds one allowance over the other, enclosing the

raw edge — shirt side seams and inseams.

2. So the upper allowance can fold completely over the lower raw edge and enclose it without bulk; untrimmed, the two allowances are equal and the fold cannot wrap the edge cleanly.
3. The binding was not centred on the edge as it fed — the ply entered the binder off-line, or the strip was hand-shifted. Re-set with the binding attachment/guide and feed the edge to the folder's centre.
4. Sample D (flat seam): it must extend with the knit. Stitch types 504 and especially 607 (cover/flatlock) provide the looped extension; 301 would crack at about 3–5% extension.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib